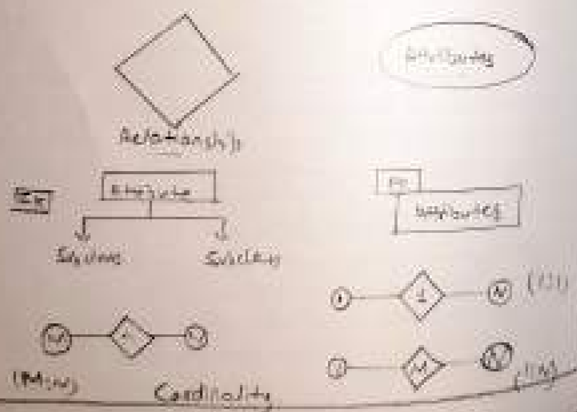


ER Diagram

Introduction :- The Entity - Relationship (ER) diagram is a conceptual data modeling technique used to visually represent the structure of a database system. It shows how data is stored, the entities involved & the relationships between them. ER diagram helps software engineers understand system requirements & design efficient database structures before implementation. They are widely used during the analysis phase of the software Development Life cycle (SDLC)

Component & Features of ER Diagram :-

(a) Component

(i) Entity -> A real world object or concept that has an independent existence.
Ex - Student, Course, Employee, Department.

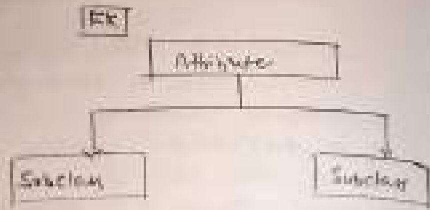
(ii) Attributes -> Characteristics or properties of an entity.
Ex Student_name, Roll_no, Course_ID.

(iii) Relationship -> Association ~~betwee~~ between two or more entities.
Ex Student enrolls in courses.

(iv) Primary Key -> Attributes that uniquely identifies each entity instance.

(v) Cardinality -> Specifies the numerical mapping between entities.

Type
- One-to-One (1:1)
- One-to-Many (1:N)
- Many-to-Many (M:N)



(xi) Foreign key → Attribute that links to primary key (PK) of another table.
Used to maintain relationship between tables.
ensure referential integrity in data base.

(xii) Weak entity → Entity that depend on another entity
Ex Order_Item depends on order.

(xiii) Derived Attributes → Attributes whose value can be calculated from another attributes.
Ex Age derived from Date_of_Birth.

Example of ER Diagram

A typical diagram for this example would show:

- Student & Course connected through Enrollment (M:N → resolve using a relationship entity).
- Instructor connected to Course with a 1:N relationship.
- Cardinality symbols near each connector
- Primary key underlined.

Viva Question

Q1 What is an ER diagram and Why is it used?

Ans ER (Entity - Relationship) diagram is a visual representation of entities, attributes & relationship in a database system.
It is used to model the structure of a database, identify how data is connected and help in designing the database during the analysis phase of software development.

Q2 Differentiate between an entity & attribute?

Ans

Entity	Attribute
A real-world object or concept that has an independent existence.	A property of characteristic the describes an entity.
Ex Student, Course, Employee.	Ex Student Name, Course ID, Employee salary.

Q3 What is the Purpose of a primary key in ER diagram?

Ans A primary key (PK) is used to uniquely identify each instance (record) of an entity.
It ensure that no two records in a table have the same value for that attribute & helps maintain data integrity.

Q4 Explain the concept of Cardinality with example.

Ans Cardinality defines the numerical relationship between two entities - how many instances of one entity relate to instances of another.

Types

- One-to-one (1:1) → One person has one passport.
- One-to-many (1:N) → One teacher teaches many student.
- Many-to-many (M:N) → Students enroll in many courses & many courses have many students.

Q5 What the difference between a weak entity & a strong entity.

Strong entity	Weak entity
Has its own primary key	Doesn't have a primary key of its own.
Exists independently	Depends on a strong entity for its existance.
Ex Student, Courses	Ex Order_item depends on Order.